1 Dynamically Sized Arrays and Amortized Running Time

Problem 1.1. Consider table expansion where we increase the size of the table by a factor of $\alpha > 1$ whenever the table is full. (For instance, a table with size 100 is expanded to size 100α. A table with size α is expanded to size α^2.) Each insertion without table expansion takes $\Theta(1)$ time. Table expansion takes cs time where s is the size of the new table. Prove that n insertions into an initially empty table of size 1 takes a total of $\Theta(n)$ time (an amortized run time of $\Theta(1)$ per insertion.)

Let $T(n)$ = Total time to insert n elements

$= [\text{Time for inserts}] + [\text{Time for expansions}]$

$$= cn + \underbrace{\sum_{k=1}^{\ell} c\alpha^k}_{\in \underbrace{\Theta(\alpha^\ell)}_{=\Theta(n)}} \quad \text{cuz } \alpha > 1.$$

old S	new S	cost
1	α	$c\alpha$
α	α^2	$c\alpha^2$
α^2	α^3	$c\alpha^3$
$\vdots$	$\vdots$	$\vdots$
$\frac{n}{\alpha}$	n	cn

$\sum_{k=1}^{\ell} c\alpha^k$

To find ℓ,

$c\alpha^\ell = cn$

$\ell = \log_\alpha(n)$

Thus, $T(n) \in \Theta(n)$

and $AT(n) \in \Theta(1)$.

Problem 1.2. Recall that amortized running time is the average time to perform a sequence of function calls. With that in mind, consider the following function

```
1: function Func(A[],n)
2:     x = 0
3:     if n is a power of 2 then
4:         for i = 1 to n^2 do
5:             x = x + A[i]
6:         end for
7:     end if
8:     for i = 1 to n do
9:         x = x + A[i]
10:    end for
11:    return x
12: end function
```

cn^2

n	cost of lines 4-6
1	c
2	$4c$
4	$16c$
$\vdots$	$\vdots$
n	cn^2

$\sum_{k=1}^{\ell} 4^k c$

$4^k c = cn^2$
$k = \log_4(n^2)$.

cn

n	cost of lines 8-10
1	c
2	$2c$
3	$3c$
$\vdots$	$\vdots$
n	cn

$\sum_{i=1}^{n} ci$

What is the asymptotic amortized running time of the following sequence of n function calls,

```
1: Func(A[], 1)
2: Func(A[], 2)
3: Func(A[], 3)
4: ...
5: Func(A[], n/2)
6: Func(A[], n/2 + 1)
7: ...
8: Func(A[], n − 1)
9: Func(A[], n)
```

That is, let $t(x)$ be the running time of $\texttt{Func}(A[], x)$, we want to find

$$T(n) = t(1) + t(2) + t(3) + \cdots + t(n/2) + t(n/2+1) + \cdots + t(n-1) + t(n)$$

Hint: This is EXTREMELY similar to table doubling.

$$T(n) = \sum_{i=1}^{n} ci + \sum_{k=1}^{\ell} 4^k c$$

$$= c\frac{n(n+1)}{2} + \underbrace{\sum_{k=1}^{\ell} 4^k c}_{\Theta(4^\ell) \text{ cuz } 4>1.}$$

$$= \Theta(n^2).$$

So, $T(n) \in \Theta(n^2)$

$AT(n) \in \Theta(n)$.